

Voltage Vector Directional Control for IPMSM Based on MTPA Strategy

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ABSTRACT Space vector pulse width modulation (SVPWM) is used in motor control because of its high dynamic response and flexible control. Maximum torque per ampere (MTPA) strategy has become the most commonly used choice for internal permanent magnet synchronous motor (IPMSM). However, complicated coordinate transformations are required in the usual SVPWM algorithm and the performance of the MTPA algorithm can be affected by the parameters variation. This paper proposes a voltage vector directional control (VVDC) method based on the MTPA strategy. The inverse Park transformation is avoided in the proposed control scheme. And the proposed algorithm not only is briefer and clearer in the sector judgment process but also has lesser calculation during the duty cycle generation. In order to realize the MTPA strategy, two voltage feedback closed loops are designed to compensate for the d -axis and q -axis currents. A simple and effective multi-parameter identification method is also presented to solve the parameters variation problem. The effectiveness of the proposed method is verified by simulation and experiments.

INDEX TERMS IPMSM, MTPA, motor parameter identification, voltage vector control.

I. INTRODUCTION

IPMSM has the advantages of high power density, high efficiency, and stable rotor structure [1]. In recent years, many scholars have studied the design and control methods of IPMSM. There are basically two major IPMSM control structures [2]: field-oriented control (FOC) and V/f control. FOC is mainly used for high-performance control applications, and SVPWM is used for decoupling control in FOC to operate the torque and the flux linkage separately. However, the usual SVPWM algorithm needs three coordinate transformations (Park, Clarke, and inverse Park transformations), which is complicated and would magnify the calculation load of the control system.

The control methods of permanent magnet synchronous motor mainly include: $i_d = 0$ control [3], constant flux linkage control [4], optimal efficiency control [5], and MTPA [6]–[8]. MTPA is an effective control method for IPMSM current regulation [6]. MTPA method utilizes the reluctance torque of IPMSM in constant torque region, which can reduce the copper loss of stator winding and can reduce the capacity of inverter relatively. However, the control effect of MTPA would become worse because of the variation of

the motor parameters. To eliminate the parameter influence on MTPA, high-frequency voltage signals are injected into the stator winding to obtain the optimal MTPA trajectory by analyzing the amplitudes of stator flux and current, but the approach is difficult to implement [1], [7]. In [8], an improved design of MTPA current control loop is presented. This method can eliminate the influence of parameters variation on control system, but the algorithm is complicated and the portability is trivial.

Rotor convex polarity, magnetic saturation, and armature reaction have a significant influence on IPMSM in different operating conditions [9]. The magnetic saturation effect of rotor core would significantly weaken flux linkage in the air gap. The d -axis and q -axis inductances would change with the amplitude and phase of phase currents. The variation of flux linkage and inductance can both influence the operation of the machine in the transient and steady state. Parameter identification methods can be effectively used to identify the motor parameters in different working conditions.

The parameter identification methods mainly include: experimental methods [10], offline identification [11], and online identification [12]. Experimental methods (such as no-load experiment, blocked rotor test, etc.) are simple, but the nonlinear characteristics of the machine in actual

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operating conditions can't be considered. Although offline identification methods have high accuracy, the signal process load is heavy. Online identification methods have been widely used in sensorless and adaptive control [13], [14]. Most of the online identification methods focus on identifying one or two parameters while keeping the other parameters fixed [15], [16]. In view of this problem, a simple and effective online multi-parameter identification method is presented in this paper.

The contribution of this paper is to propose a method of VVDC method based on the MTPA strategy. In the proposed method, the inverse Park transformation is avoided. The VVDC not only is briefer and clearer in the sector judgment process but also has lesser calculation. A simple and effective identification method based on the recursive least squares (RLS) algorithm is presented to acquire the motor parameters. These parameters are used to calculate the optimal MTPA trajectory.

The rest of this paper is organized as follows: first, the relationship between the amplitude and the phase of the expected voltage vector in the MTPA strategy is illustrated in Section II. To eliminate the influence of the parameters variation on MTPA, a simple and effective online multi-parameter identification method is then presented in Section III. Section IV shows the proposed control scheme. In section V, the effectiveness of the proposed method is verified by simulation and experiments. Finally, the conclusion of this paper is given in Section VI.

II. MATHEMATICAL MODEL OF IPMSM

Neglecting the copper and iron losses, magnet motive force harmonics of windings, and structural asymmetry, the standard diagram of a salient synchronous machine at a steady state is shown in Fig.1. The mathematical model of IPMSM defined in the rotating reference frame d - q can be expressed through (1) - (4) [17], [18].

$$V_s \sin \delta = X_q I_s \cos \psi + R I_s \sin \psi \quad (1)$$

$$V_s \cos \delta = E_o - X_d I_s \sin \psi + R I_s \cos \psi \quad (2)$$

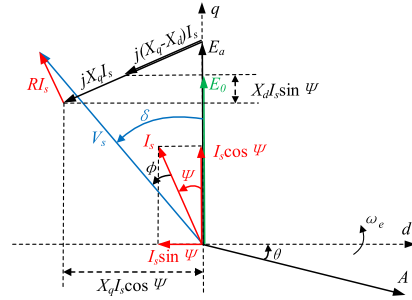
$$E_o = 2\pi f_e \lambda_m \quad (3)$$

$$\psi = \delta - \phi \quad (4)$$

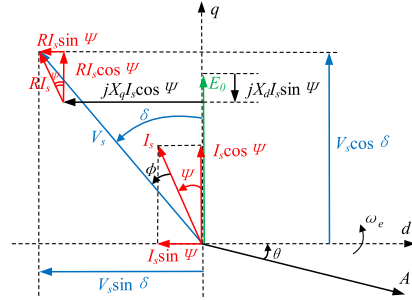
where V_s and I_s are the phase stator voltage and phase stator current, X_d and X_q are the reactance in the d - q reference frame. δ is the load angle, ψ is the angle between I_s and q -axis, ϕ is the phase angle between V_s and I_s . R , λ_m , f_e , and E_o are armature winding resistance, permanent-magnet flux linkage, electrical frequency, open-circuit voltage, respectively.

From (1), the relationship between the angle δ and ψ can be expressed as

$$\delta = \arcsin\left(\frac{X_q I_s \cos \psi + R I_s \sin \psi}{V_s}\right) \quad (5)$$



(a) The relation diagram of the phasor.



(b) The decomposition diagram of the phasor.

FIGURE 1. The phasor diagram of IPMSM.

where V_s can be calculated from equations (1) and (2), as given in (6).

$$V_s = \sqrt{(E_o - X_d I_s \sin \psi + R I_s \cos \psi)^2 + (X_q I_s \cos \psi + R I_s \sin \psi)^2} \quad (6)$$

In equations (5) and (6), the values of δ and V_s can be regarded as the function of ψ . The two equations are both valid under any steady-state working conditions [17]. Therefore, the phase angle and amplitude of the stator voltage can be controlled by calculating equations (5) and (6) respectively.

In the MTPA strategy, the stator current amplitude is expected to be minimized under different load conditions for producing maximum torque per ampere. When there is no rotor saliency, MTPA is achieved by aligning the phase stator current vector with the q -axes. For IPMSM, i.e., $X_d < X_q$, rotor saliency makes such a strategy performed by a suitable profile of the combination of the d - q axes currents. This combination can be expressed through the following expressions [19], [20]

$$I_s \sin \psi = \frac{\lambda_m - \sqrt{\lambda_m^2 + 8(L_d - L_q)^2 I_s^2}}{4(L_d - L_q)} \quad (7)$$

$$I_s \cos \psi = \sqrt{I_s^2 - (I_s \sin \psi)^2} \quad (8)$$

where L_d and L_q are d - and q -axis stator inductances.

Equation (7) allows calculating the angle ψ

$$\psi = \arcsin\left(\frac{\lambda_m - \sqrt{\lambda_m^2 + 8(L_d - L_q)^2 I_s^2}}{4(L_d - L_q) I_s}\right) \quad (9)$$

By substituting (9) into (5) and (6), the expressions of V_{s_opt} and δ_{opt} in the MTPA strategy are obtained

$$V_{s_opt} = \sqrt{\frac{(E_o - X_d I_s \frac{\lambda_m - A}{4B} + R I_s \frac{4B^2 - \lambda_m^2 + \lambda_m A}{2B})^2}{(X_q I_s \frac{4B^2 - \lambda_m^2 + \lambda_m A}{2B} + R I_s \frac{\lambda_m - A}{4B})^2}} \quad (10)$$

$$\delta_{opt} = \arcsin \left[\frac{X_q I_s (8B^2 - 2\lambda_m^2 + 2\lambda_m A) + R I_s (\lambda_m - A)}{4B V_s} \right] \quad (11)$$

where the expressions of A and B are given by

$$A = \sqrt{\lambda_m^2 + 8(L_d - L_q)^2 I_s^2}, \quad B = (L_d - L_q) I_s$$

It can be seen from (10) and (11), in a high-performance MTPA strategy, the machine parameters R , λ_m , L_d and L_q are required. The precise knowledge of these parameters is important to obtain the best control performance. Parameters online identification can be used to obtain the motor parameters under different working conditions.

III. MULTI-PARAMETER IDENTIFICATION

From the steady-state mathematical model of IPMSM, as shown in (1) and (2), the voltage equation in d - q coordinates can be rewritten as

$$v_d = R i_d - \omega_e L_q i_q \quad (12)$$

$$v_q = R i_q + \omega_e L_d i_d + \omega_e \lambda_m \quad (13)$$

where v_d , v_q , and i_d , i_q are the stator voltages and currents in d -axis and q -axis respectively, ω_e is electrical angular velocity. v_d and v_q are obtained from the two PI regulators.

In this paper, the initial values of phase resistance and magnet flux linkage are determined by experimental tests. The d - q axis voltages and currents can be obtained by current sensors through coordinate transformation.

Assuming that the skin loss versus the stray loss of the winding are negligible, the phase resistance R is a linear function of winding temperature [16], then the actual value of phase resistance at temperature T_1 is given by

$$R = R_0 + \alpha(T_1 - T_0) \quad (14)$$

where R_0 is the initial value of phase resistance at room temperature ($T_0 = 22^\circ\text{C}$), α is the temperature coefficient, T_1 is the actual temperature measured by the temperature sensor.

From (12) and (13), the values of L_d and L_q can be calculated by (15) and (16).

$$L_d = \frac{v_q - R i_q - \omega_e \lambda_m}{\omega_e i_d} \quad (15)$$

$$L_q = \frac{R i_d - v_d}{\omega_e i_q} \quad (16)$$

By using the identified values of R and L_d , the value of λ_m can be calculated from (17).

$$\lambda_m = \frac{v_q - R i_q - \omega_e L_d i_d}{\omega_e} \quad (17)$$

Owing to attractive advantages of RLS algorithm (such as simplicity, good stability, and strong robustness), it has been commonly used to solve the identification problems. RLS algorithm with forgetting factor is adopted in this paper to identify parameters from (15), (16), and (17), respectively. The equations of the RLS algorithm with a forgetting factor are given in (18).

$$\begin{cases} \varepsilon_N = y_N - \varphi_N^T \eta_{N-1} \\ \eta_N = \eta_{N-1} + K_N \varepsilon_N \\ K_N = \frac{P_{N-1} \varphi_N}{\lambda I + \varphi_N^T P_{N-1} \varphi_N} \\ P_N = \frac{1}{\lambda} (P_{N-1} - K_N \varphi_N^T P_{N-1}) \end{cases} \quad N = 1, 2, 3 \dots \quad (18)$$

where ε_N is the estimation error, y is the output matrix, φ is the feedback matrix, η is the estimated parameter vector, λ is the forgetting factor, I is the identity matrix, K and P are the update gain matrices. The value of the forgetting factor is usually chosen between 0.9 and 1 [21]. In this work, λ is set as 0.95.

Based on equations (12), (13) and (18), L_d and L_q can be identified at the first step. The RLS estimation equation of L_d and L_q is described by (19).

$$\begin{cases} y = \begin{bmatrix} R i_d - v_d \\ v_q - R i_q - \omega_e \lambda_m \end{bmatrix} \\ \varphi^T = \begin{bmatrix} 0 & \omega_e i_q \\ \omega_e i_d & 0 \end{bmatrix} \\ \eta = \begin{bmatrix} L_d \\ L_q \end{bmatrix} \end{cases} \quad (19)$$

The identified value of L_d can be used in the next identification process. The RLS estimation equation of λ_m can be described by (20).

$$\begin{cases} y = [v_q - R i_q - \omega_e L_d i_d] \\ \varphi^T = [\omega_e] \\ \eta = [\lambda_m] \end{cases} \quad (20)$$

The multi-parameter identification structure of the employed RLS algorithm with a forgetting factor is shown in Fig. 2. By using the multi-parameter identify method, the machine parameters can be identified. Then, these identified parameters can be used to calculate the value of V_{s_opt} and δ_{opt} .

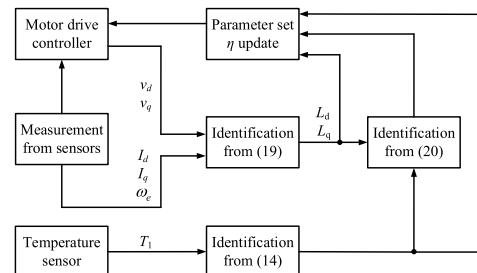


FIGURE 2. Multi-parameter identification block diagram.

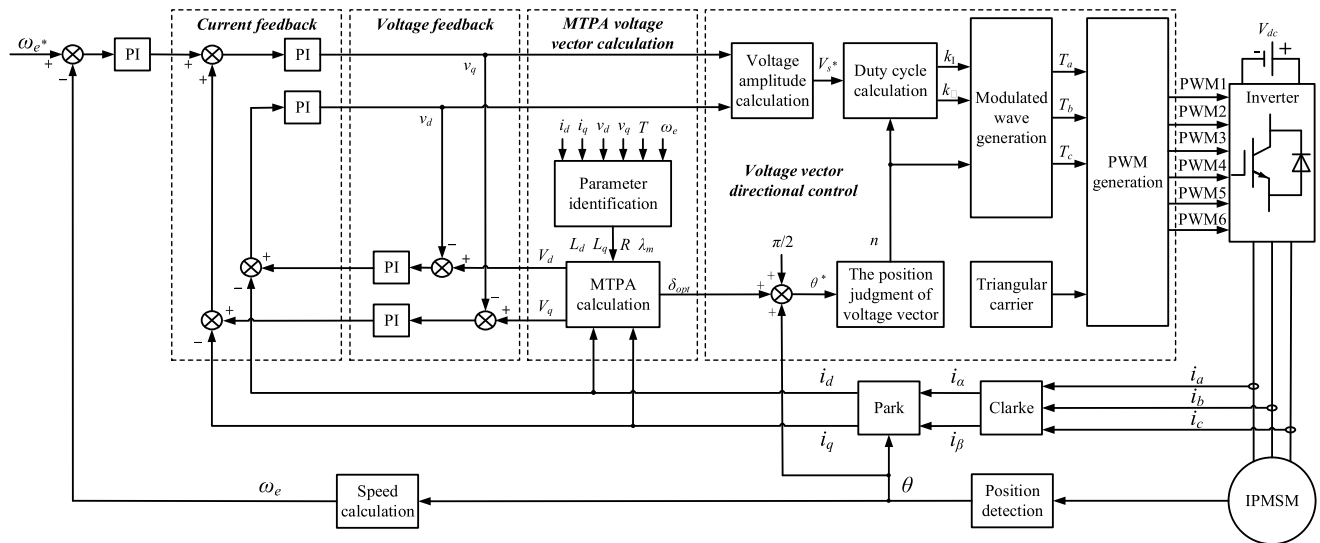


FIGURE 3. Block diagram of the proposed control scheme.

IV. MTPA VOLTAGE VECTOR DIRECTIONAL CONTROL

A. INTEGRATION OF MTPA STRATEGY

Fig. 3 shows the block diagram of the proposed VVDC scheme based on MTPA strategy. Essentially, to realize MTPA, two voltage feedback closed loops are designed to compensate the d -axis and q -axis currents. Note that the regulation of voltages by using the two proportional-integral (PI) regulators in the voltage feedback closed loops is the tool to achieve MTPA. In Fig. 3, the inverse Park transformation is avoided, which can make the control algorithm be simplified.

The reference values of d - and q - axis stator voltages can be calculated from (21) and (22) respectively.

$$V_d = -V_{s \text{ opt}} \sin \delta_{opt} \quad (21)$$

$$V_q = V_{s_opt} \cos \delta_{opt} \quad (22)$$

In equations (10) and (11), the calculation of V_{s_opt} and δ_{opt} still requires motor parameters and I_s . The parameters can be obtained by the multi-parameter identification method, as shown in Fig.2. The calculation of I_s is given in (23).

$$I_s = \sqrt{i_d^2 + i_q^2} \quad (23)$$

B. SECTOR JUDGMENT

In IPMSM field-oriented control, as shown in Fig. 3, this scheme is implemented by generating the gate pulse which is needed by the inverter. There are eight switching states for the three-phase inverter in a - b - c stationary frame (i.e. 000, 001, 010, 011, 100, 101, 110, 111, where 0 means the lower arm is conducted and 1 means the upper arm is turned on). These states correspond to eight basic vectors ($V_0 \sim V_7$), as shown in Fig. 4. V_0 and V_7 (not shown) are zero vectors located at the coordinate origin o [22].

There are six sectors in Fig. 4, the sector number increases counterclockwise from I to VI. The position of V_{s_opt} can be

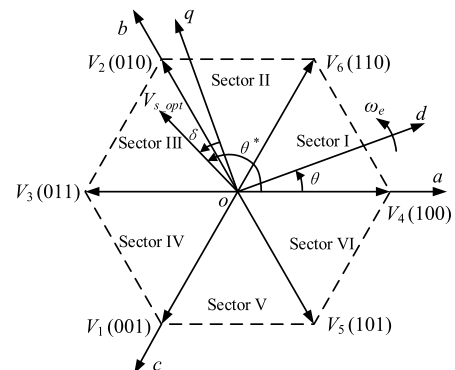


FIGURE 4. Voltage vector position diagram.

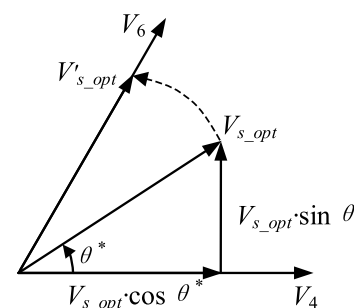


FIGURE 5. V_{s_opt} located in the sector I.

expressed as

$$\theta^* = \theta + \delta_{opt} + \frac{\pi}{2} \quad (24)$$

where θ is the angle between the rotor flux direction and the a -axis, θ^* is the angle between V_{s_opt} and the a -axis.

The sector in which V_{s_opt} is located can be determined by θ^* . Assuming that V_{s_opt} is located in the sector I, the diagram can be seen in Fig. 5.

As shown in Fig. 5, θ^* is ranged from $0 \sim \pi/3$. When aligning V_{s_opt} with V_6 , i.e. V_{s_opt} reaches V'_{s_opt} , the maximum value of $\sin \theta^*$ can be obtained; when V_{s_opt} coincides with V_4 , $\sin \theta^*$ achieves the minimum value. In order to ensure that V_{s_opt} is located in the sector I, $\cos \theta^* > 0$ should also be satisfied. The judgment condition that V_{s_opt} located in the sector I is given by

$$\begin{cases} 0 < \sin \theta^* < \sqrt{3}/2 \\ 0 < \cos \theta^* \end{cases} \quad (25)$$

The sector judgment value can be given as 1 when V_{s_opt} is located in the sector I. Similarly, the judgment conditions of the six sectors can be obtained, as shown in Table 1.

C. DUTY CYCLE CALCULATION

The two basic vectors can be selected according to the sector in which the desired voltage vector is located. Assuming that the expected voltage V_s^* is located in the sector I, and V_s^* can be decomposed by adopting the parallelogram law, as shown in Fig. 6.

By means of the sine theorem, the relationship between the two basic vectors and V_s^* is given in (26).

$$\frac{V_s^*}{\sin 120^\circ} = \frac{k_4 V_4}{\sin(60^\circ - \theta^*)} = \frac{k_6 V_6}{\sin \theta^*} \quad (26)$$

where $V_4 = V_6 = 2V_{dc}/3$ (V_{dc} is the bus voltage), k_4 and k_6 are the duty cycles of V_4 and V_6 , respectively. The magnitude

TABLE 1. Judgment condition of voltage vector position.

Sector	Judgment condition	Judgment value
I	$0 < \sin \theta^* < \sqrt{3}/2, \cos \theta^* > 0$	1
II	$\sin \theta^* > \sqrt{3}/2$	2
III	$0 < \sin \theta^* < \sqrt{3}/2, \cos \theta^* < 0$	3
IV	$-\sqrt{3}/2 < \sin \theta^* < 0, \cos \theta^* < 0$	4
V	$\sin \theta^* < -\sqrt{3}/2$	5
VI	$-\sqrt{3}/2 < \sin \theta^* < 0, \cos \theta^* > 0$	6

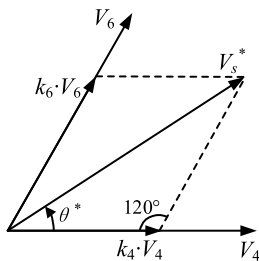


FIGURE 6. The decomposition diagram.

of the voltage vector V_s^* can be calculated by

$$V_s^* = \sqrt{v_d^2 + v_q^2} \quad (27)$$

From (26) and (27), k_4 and k_6 can be expressed as

$$k_4 = \frac{\sqrt{3 \cdot (v_d^2 + v_q^2)} \sin(60^\circ - \theta^*)}{V_{dc}} \quad (28)$$

$$k_6 = \frac{\sqrt{3 \cdot (v_d^2 + v_q^2)} \sin \theta^*}{V_{dc}} \quad (29)$$

Similarly, the duty cycle expressions of the other five sectors can be obtained. The duty cycle expressions can be uniformly represented by the following relationship

$$k_I = \frac{\sqrt{3 \cdot (v_d^2 + v_q^2)} \sin(n \cdot 60^\circ - \theta^*)}{V_{dc}} \quad (30)$$

$$k_{II} = \frac{\sqrt{3 \cdot (v_d^2 + v_q^2)} \sin(\theta^* - (n-1) \cdot 60^\circ)}{V_{dc}} \quad (31)$$

where n represents the sector judgment value ($n = 1, 2, 3 \dots 6$), k_I and k_{II} represent the duty cycle of the two basic voltage vectors (counterclockwise), respectively.

D. MODULATED WAVE GENERATION

After the duty cycle is obtained, the modulated waves T_A , T_B , and T_C can be calculated based on the sector judgment value. The generation method in other different sectors is shown in Fig. 7.

Assuming that the expected voltage is located in the sector I, the PWM generation method is shown in Fig. 8. Where T_s is switching period, $T_m = k_m T_s$ ($m = 0, 1, 2 \dots 7$).

From Fig. 8, T_A , T_B , and T_C are available to be obtained by

$$\begin{cases} T_A = (k_I + k_{II} + k_0) \cdot T_s \\ T_B = (k_{II} + k_0) \cdot T_s \\ T_C = k_0 \cdot T_s \end{cases} \quad (32)$$

where $k_I = k_4$, $k_{II} = k_6$, $k_0 = (1 - k_I - k_{II})/2$.

Similarly, the expressions of T_A , T_B , and T_C in the other five sectors can be obtained. And the proposed modulation method is given in Table 2. Finally, the PWM pulses can

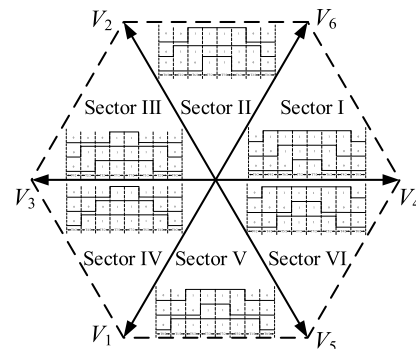


FIGURE 7. The generation method of modulated waves in different sectors.

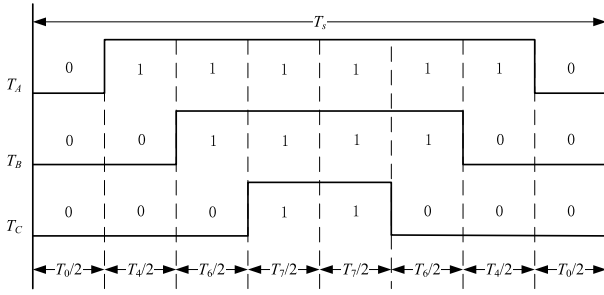


FIGURE 8. PWM generation method in the sector I.

TABLE 2. The calculation of T_A , T_B , and T_C in other sectors.

Sector	The Expressions of T_A , T_B , AND T_C
II	$\begin{cases} T_A = (k_I + k_0) \cdot T_s \\ T_B = (k_I + k_{II} + k_0) \cdot T_s \\ T_C = k_0 \cdot T_s \end{cases}$
III	$\begin{cases} T_A = k_0 \cdot T_s \\ T_B = (k_I + k_{II} + k_0) \cdot T_s \\ T_C = (k_{II} + k_0) \cdot T_s \end{cases}$
IV	$\begin{cases} T_A = k_0 \cdot T_s \\ T_B = (k_I + k_0) \cdot T_s \\ T_C = (k_I + k_{II} + k_0) \cdot T_s \end{cases}$
V	$\begin{cases} T_A = (k_{II} + k_0) \cdot T_s \\ T_B = k_0 \cdot T_s \\ T_C = (k_I + k_{II} + k_0) \cdot T_s \end{cases}$
VI	$\begin{cases} T_A = (k_I + k_{II} + k_0) \cdot T_s \\ T_B = k_0 \cdot T_s \\ T_C = (k_I + k_0) \cdot T_s \end{cases}$

be obtained by comparing the modulated wave with the triangular carrier [21], [22].

V. SIMULATION AND EXPERIMENTAL RESULTS

Simulation and experiments are conducted to verify the effectiveness of the proposed control method. The IPMSM is connected to a three-phase voltage source inverter, and the switching frequency is fixed at 10kHz. The same IPMSM is used for simulation and experiments. The motor parameters are given in Table 3.

A. SIMULATIONAL RESULTS

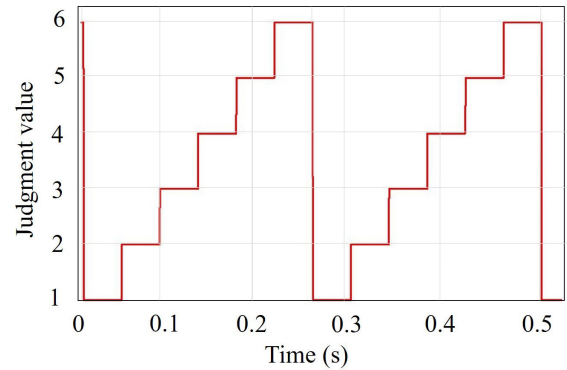
The simulation is performed in MATLAB/Simulink. Fig. 9 shows the simulation results of the sector judgment.

In Fig. 9 (a), the judgment value changed from 1 to 6 periodically, and this trend is consistent with the change of the sector numbers (as shown in 4, from I to VI). The comparison between Fig. 9 (a) and (b) is given in Table 4.

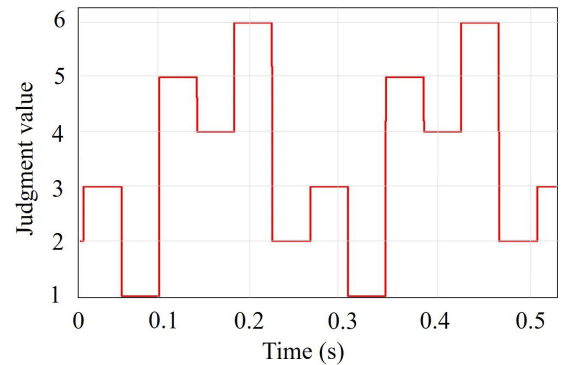
As presented in Table 4, the proposed sector judgment method is more straightforward because the judgment value is matched with the sector, while in the usual SVPWM,

TABLE 3. The parameters of IPMSM.

Parameter	Value
Rated Power	90kW
Rated speed	2100r/min
Rated current	300A
Armature winding resistance	0.03Ω
Inductance in d -axis	0.59mH
Inductance in q -axis	2.85mH
Flux linkage	0.092Wb



(a) Sector judgment value of the proposed method.



(b) Sector judgment value of the usual SVPWM.

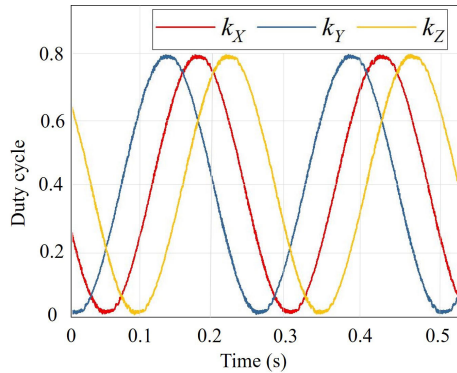
FIGURE 9. The simulation results of the sector judgment.

the judgment value should have corresponded to the sector number [23]. Therefore, in the sector judgment process, the proposed algorithm is briefer and clearer than the usual SVPWM.

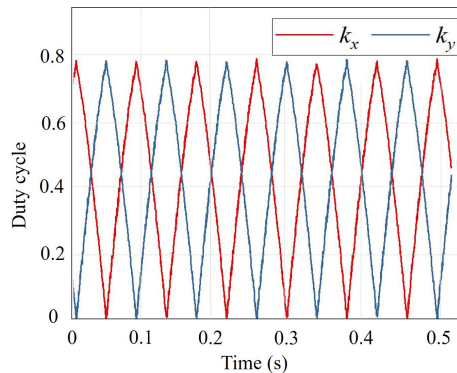
In the usual SVPWM algorithm, the generation of modulation waves can be divided into three steps [24], [25]. First, the duty cycle of the three basic voltage vectors (k_X , k_Y , and k_Z) should be calculated, as shown in Fig. 10 (a). Two voltage vectors (k_X and k_Y) are then selected based on the sector in which the expected voltage located. The duty cycle simulation result of the two selected vectors is shown in Fig. 10 (b).

TABLE 4. The sector and sector judgment value.

Sector	I	II	III	IV	V	VI
Judgment value of the proposed method	1	2	3	4	5	6
Judgment value of the usual SVPWM	3	1	5	4	6	2



(a) The duty cycle calculation of the three basic voltage vectors.



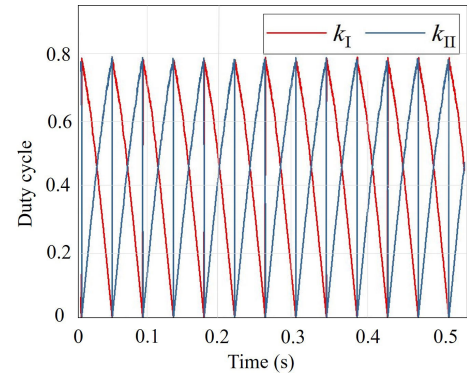
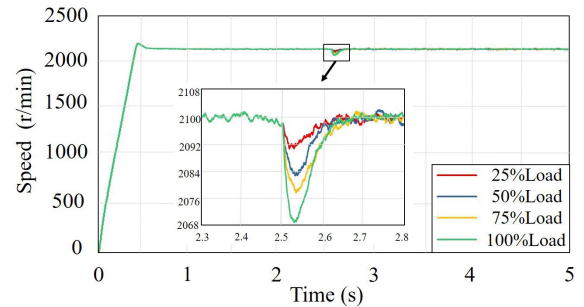
(b) The duty cycle calculation of the two selected voltage vectors.

FIGURE 10. The duty cycle calculation of the usual SVPWM algorithm.

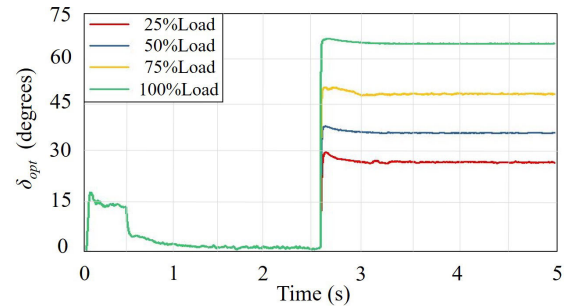
Finally, the modulation wave can be obtained according to the two duty cycles and the sector through the adoption of the specific modulation method. As shown in Fig. 10, the duty cycle calculation process of the usual SVPWM is complexed because two steps should be employed.

The duty cycle calculation result of the proposed VVDC method is shown in Fig. 11. In the proposed algorithm, the duty cycle of the two selected voltage vectors can be obtained directly without calculating the duty cycle of the three basic voltage vectors. Therefore, the duty cycle calculation method used in this paper has less calculation because two duty cycles (k_I and k_{II}) are involved, while the other three extra duty cycles are needed to be calculated in the usual SVPWM method.

After k_I and k_{II} are calculated, the modulation wave can be generated by using the proposed modulation method (as shown in Table 2). The PWM can be obtained by

**FIGURE 11.** The duty cycle calculation of the proposed method.

(a) The simulation result of speed.

(b) The simulation result of δ_{opt} .**FIGURE 12.** Transient and steady-state simulation results of the proposed method.

comparing the modulation wave with the triangular carrier wave, and the PWM is then worked to the inverter.

The transient and steady-state simulation results of the proposed method is shown in Fig. 12. The rated speed is set as the target, and different loads are applied at 2.5 s. In Fig. 12 (a), the steady-state of the motor can be reached at 0.6 s, and the speed fluctuates in a small range around 2100 r/min. During the transient, the speed decreases temporarily when the load is given at 2.5 s. And the speed reaches to the rated speed again at 2.7 s. The speed simulation result shows that the loading process of the proposed method can be completed in 0.2 s and the speed regulation performance is good. The simulation result of δ_{opt} is shown in Fig. 12 (b). The value of δ_{opt} is nearly to be zero in no-load condition, and the value would change immediately after the load is applied at 2.5 s.

After 2.7 s, the output of δ_{opt} is stable in the steady-state, and the value of δ_{opt} increases with the load.

B. EXPERIMENTAL RESULTS

For evaluation of the proposed control strategy, some tests have been carried out. The experimental setup is shown in Fig. 13.

The experimental results of A-phase current and A-B line voltage are shown in Fig. 14. The waveforms are shown in Fig. 14 (a), where the current waveform is a sine wave, and the voltage waveform is composed of high-frequency pulses. The total harmonic distortion (THD) analysis results of the current and the voltage waveforms are shown in Fig. 14 (b) and (c), respectively. The comparison of THD results between the proposed method and the usual SVPWM is given in Table 5.

As can be seen in Table 5, the THD results of the current and the voltage of the proposed method are 6.57% and 167.62%, respectively. And these percentages are consistent with the THD results of the usual SVPWM. Therefore, same control effects can be achieved by employing the proposed method.

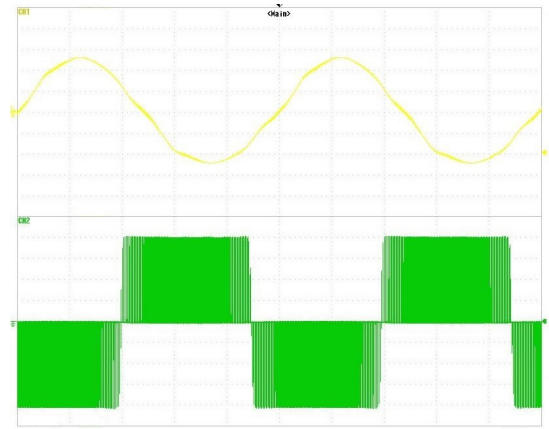
The motor parameters under different load conditions should be obtained in the first step. By using the proposed multi-parameter identify method, the parameters identified results of L_d , L_q , and λ_m are shown in Fig. 15.

As shown in Fig. 15, the value of L_d would decrease rapidly with the increase of d and q axis currents. And the maximum value of L_d is almost twice as high as the minimum. The values of L_q and λ_m are not sensitive to d axis current. However, the q axis current both has a notable impact on L_q and λ_m . As can be seen in Fig. 15, these motor parameters change

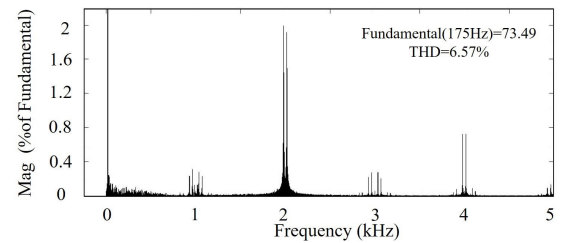
in a wide range with respect to the $d - q$ axis currents. The parameter variation would have a significant influence on the MTPA trajectory.

The loading test is performed for evaluating the feasibility of the proposed VVDC method and verifying the effectiveness of parameter identification. The proposed method with parameters identification (proposed MTPA) and the proposed method without parameters identification (constant-parameter MTPA) are compared in Fig. 16.

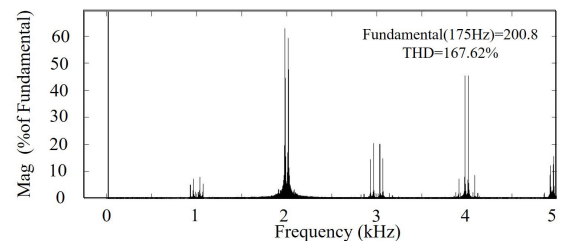
As shown in Fig. 16, the load is given at 0.5 s and reached 100 N·m at 2.8 s. From the current experimental results in Fig 16 (a) and (b), the amplitude of the phase current increases at 0.5 s gradually, and reaches a stable state at 2.8 s. The proposed VVDC method can regulate the motor effectively. In Fig 16 (a) and (b), the amplitude of the phase current is 142 A and 154 A, respectively. Therefore, in the same load condition, the current amplitude of the proposed method is smaller, which would reduce the copper consumption and improve the efficiency of the motor [26]. Fig 16 (c) shows the result of δ_{opt} in the loading process. The value of



(a) The waveform of the phase current and line voltage. (CH1) A-phase current (30A/div). (CH2) A-B line voltage (50V/div). Time: 2ms/div.



(b) THD analysis of the phase current.



(c) THD analysis of the line voltage.

TABLE 5. The comparison of the THD results.

Method	Current THD	Voltage THD
Proposed method	6.57%	167.62%
Usual SVPWM	6.59%	167.65%

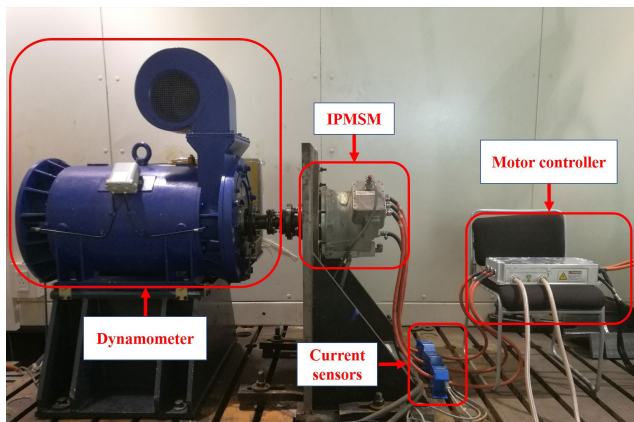


FIGURE 13. Experimental setup.

FIGURE 14. Experimental results of the phase current and line voltage.

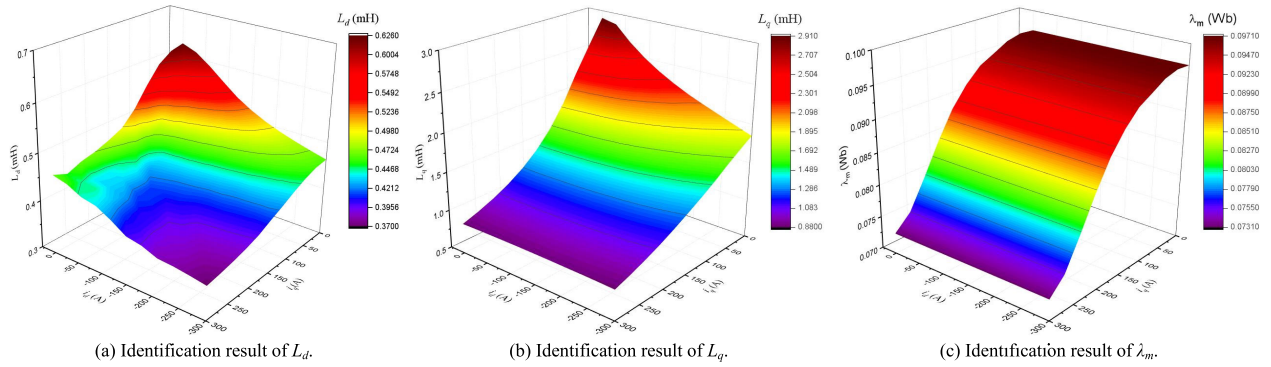


FIGURE 15. The experimental results of parameters identification.

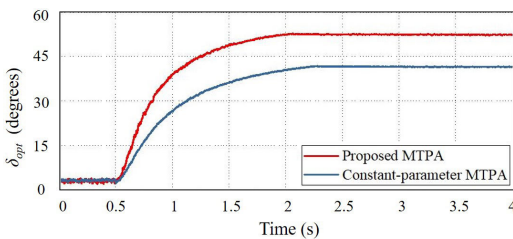
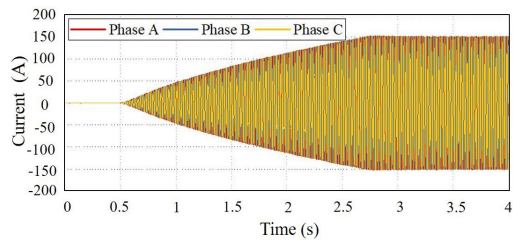
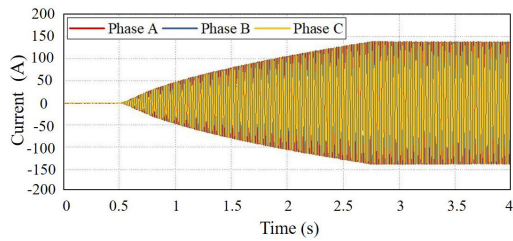


FIGURE 16. The results of the loading test.

δ_{opt} is nearly to be zero in no-load condition, and the value increases gradually after the load is given at 0.5 s. The δ_{opt} can reach a stable value in the steady-state. As can be seen from Fig 16 (c), the value of δ_{opt} calculated by the proposed method is 53 degrees, while the value of the constant-parameter MTPA is 42 degrees. The value of δ_{opt} would lead to a different performance on the MTPA trajectory. And this is the reason that the proposed method has a smaller amplitude of the phase current.

The results of δ_{opt} at different loads conditions is presented in Fig 17. As can be seen, the value of δ_{opt} increases with the stator current. The δ_{opt} obtained by the proposed method

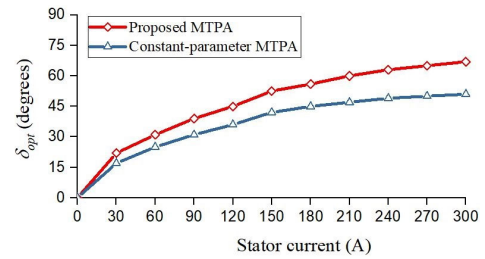


FIGURE 17. The results of δ_{opt} at different loads.

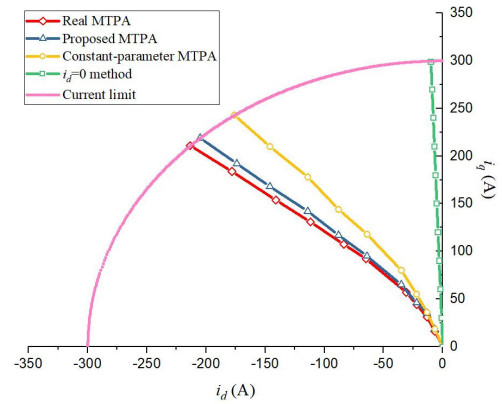


FIGURE 18. Experimental current trajectories of different control strategies.

is higher than the constant-parameter MTPA method under different loads. This illustrates that the δ_{opt} acquired by the proposed method would affect the MTPA trajectory under different loads.

Three different kinds of control strategies are compared in this paper. The experimental comparison among the proposed MTPA, constant-parameter MTPA, and $i_d = 0$ control method is conducted. The experimental results and the real MTPA trajectory are shown in Fig. 18.

In Fig. 18, the trajectory of $i_d = 0$ method approaches i_q -axis. The trajectory of constant-parameter MTPA is inconsistent with its corresponding real MTPA trajectory for neglecting the parameters variation. The proposed MTPA trajectory agrees well with the real MTPA trajectory.

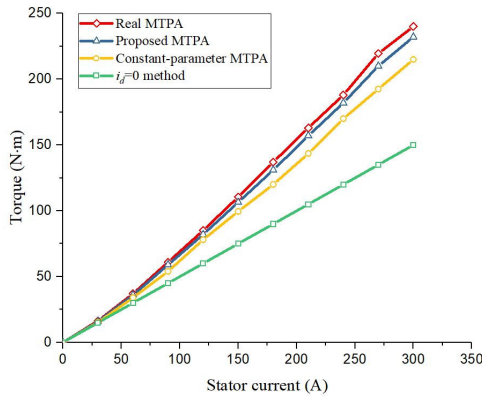


FIGURE 19. The relationship between stator current and torque of different control strategies.

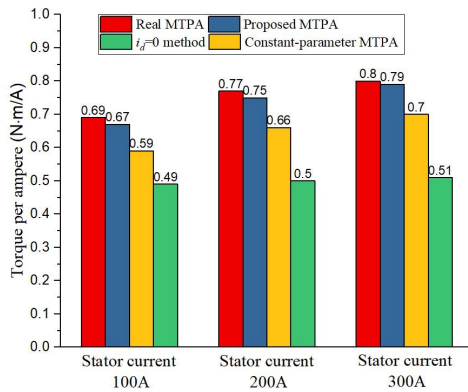


FIGURE 20. Torque per ampere of different control methods.

TABLE 6. The proportions of the three control methods.

Stator current (A)	Proposed MTPA	Constant-parameter MTPA	$i_d=0$ method
100	97.1%	85.5%	71.0%
200	97.4%	85.7%	64.9%
300	98.8%	87.5%	63.8%
Average value	97.77%	86.23%	66.57%

The relationship between stator current and torque of different control strategies are shown in Fig. 19.

In Fig. 19, the proposed MTPA method has higher output torque at the same stator current. The experimental results of the three control methods under different stator currents are shown in Fig. 20.

As shown in Fig. 20, the torque per ampere value of the $i_d = 0$ method keeps nearly constant while the values of the others increase along with the stator current. The result of the proposed MTPA method approaches to the real MTPA. Compared with real MTPA trajectory (considered as 100%), the proportions of the three control methods are shown in Table 6.

As can be seen from Table 6, the average proportions of the three methods are 97.77%, 86.23%, and 66.57%, respectively. Compared with the other two methods, the proposed MTPA method can respectively increase the value of torque per ampere by 11.54% and 31.2% on average.

VI. CONCLUSION

SVPWM algorithm and MTPA control strategy are commonly used in IPMSM control. In order to simplify the control algorithm and eliminate the influence of parameters variation on MTPA, this paper has proposed a novel VVDC method based on MPAT strategy. Compared with the usual SVPWM, the proposed VVDC algorithm is briefer and clearer in the process of sector judgment, and has lesser calculation during the duty cycle calculating process. Moreover, the inverse Park transformation is avoided which can make the control algorithm become more simplified. For realizing the MTPA strategy, two voltage feedback closed loops are designed to compensate the d -axis and q -axis currents. The machine parameters, which identified by the presented online multi-parameter identification method, have been used in the calculation of MTPA trajectory. The effectiveness of VVDC method based on MTPV strategy has been verified by simulation and experiments. The experimental results prove that the accuracy of the proposed control method can achieve 97.77% of the real MTPA trajectory.

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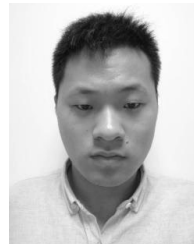


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parameter identification of the permanent magnet synchronous motor.

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